Fiscal Year:	FY 2007	**Task Last Updated:**	FY 11/10/2006
PI Name:	German, Randall M. Ph.D.		
Project Title:	Gravitational Effects on Distortion in Sintering		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	92014	**Congressional District:**	53
Comments:	co-investigator Eugene A Olevsky, SDSU, Elisa Torresani, SDSU, MS 1326		
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	02/01/1998	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2006 per PI information (11/06)		
Key Personnel Changes/Previous PI:	2		
COI Name (Institution):			
Grant/Contract No.:	NAG8-1452		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		

Task Description:	Particle-based fabrication studies using a viscous or liquid phase for particle bonding with low-gravity conditions that impact densification and distortion. These studies are performed using systematic variations of the controlling parameters of porosity, solid:liquid ratio, and basic thermodynamic features such as solubility and dihedral angle. Further, to examine the role of porosity and pore size (and associated buoyancy effcts) on sintering densification and distortion, with attention to formation of large pore agglomerates, possible distortion due to pore buoyancy, relations between distortion and densification, and assessment of capillary effects on both. Calculation and measurement of in situ strength and viscosity. Use of the evolving mechanical property data to feed into finite element and fluid dynamic calculations to predict component size, shape, and defects during sintering densification. To support this model, quenching experiments are being used to freeze the evolving microstructures to measure density and pore saturation for identification of possible options to resist distortion during sinteirng. Extraction of gravity effect is important since there three forces causing distortion during densification - anisotropic gravity forces, and hydrostatic forces from capillarity, and density (porosity and pore size) gradients. One of the findings is that we can now manipulate the sintering cycle to improve strength and avoid distortion, a finding that relates to simple changes in thermal cycle design and should therefore have wide industrial appeal. Full three dimensional computer simulations in a range of gravitational conditons are a recent task emphasis. 1) Investigate the effect of porosity on the microstructure and compact shape retention during sintering in the presence of a liquid phase, with separation of gravitational role via pore buoyancy. 2) Systematically and independently vary the pore size and porosity level and location to assess pore effects on distortion or slumping during liquid phase sintering while examining a range of dihedral angles, solid:liquid ratios, and sintering time effects. 3) Investigate buoyancy-induced pore migration as part of the densification process to determine if vacancy annhilation is the only densification process; further evaluate pore migration effects on induced distortion. 4) Time resolved determination of the sequence of events and interaction of events leading to distortion and densification, thereby extracting any possible separation of the two events for practical benefits. 5) Determine if there are unique thermal strategies to avoid distortion during sintering, especially during the periods of low strength as isolated in the experiments and modeling. 6) Demonstration of scaling laws to model proposed systems in various gravitational environments to improve dimensional precision and to predict final size, shape, and properties. This is extended to a range of gravitational conditions including earth, mars, lunar, and space. Fundamentally, sintering is an important net-shaping technology applied to metals, ceramics, and cermets, composites, cemented carbides and other mixed powder systems. Sintering approaches are a favorite to freeform fabrication routes, many of which are being considerd for in-space fabrication and repair. Further, recent efforts have suggested sintering as a means to use space-based materials for fabrication of living, launch, and other structures. In the USA, sintered materials contribute over $23 billion to the economy. This research is applicable to improved sintering cycles and the production of higher quality, more precise components. It combines intentional green body changes in composition, porosity, pore size, and liquid content with the presence or absence of gravity to develop our understanding in this technologically significant area. The source of component distortion in sintering and the time-dependent role of porosity are poorly understood in sintering practice. Pores are very stable under microgravity conditions. Consequently, microgravity provides the unique opportunity to control and manipulate pores to extract the role of porosity and pore size with respect to densification and distortion. Further, changes in coposition and thermal cycle are being used to improve the component rigidity (strength) at temperature via both modeling and experimental efforts. We have determined that nondestructive tests can now possibly provide needed information for thermal cycle design to minimize distortion. Further, tool or green body designs are now possible based on specified final size, shape and density.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has implications in all aspects of human activity on earth. For example, polar ice caps contain trapped atmospheres from past times, providing a means to determin the greenhouse changes if samples can be recovered and dated. Sintering models used in the current project have been trained for ice (snow) sintering to provide a calculation as to the time when pores closed versus depth (gravitational head) and annual snowfall. Likewise, the models developed in this work show how mankind can fabricate precise components, thereby minimizing wasted materials and efforts in the fabrication of industrial compnents. With respect to the burden of space launch, this work provides fundamental data on the role of gravity (and microgravity) with respect to various proposed in-space fabrication and repair routes. The cost of launching various freeform fabrication devices is enormous, both in direct cost and in space operating difficulties. This work is developing accurate models for final size and shape that can be mapped onto the operating parameters of various proposed freeform fabrication and repair ideas to show realistic feasiblity, without the burden of launch.
Task Progress:	With termination of funding, the effort has been totally focused on degree completion and the last MS student will graduate this winter. At that point all materials, samples, records, and findings will be archived for possible future use in publications and any use for books, presentations, or other uses. A few presentations were made on the summary findings.
Bibliography Type:	Description: (Last Updated: 03/27/2025)
Abstracts for Journals and Proceedings	Campbell LG, German RM. "Gravitational Effects on Hardness Liquid Phase Sintering." International Conference On Powder Metallurgy & Particulate Materials, Denver, Colorado, May 13–16, 2007. In: Advances in Powder Metallurgy & Particulate Materials—2007 Proceedings of the 2007 International Conference On Powder Metallurgy & Particulate Materials, May 13–16, 2007, Denver, Colorado. Princeton, NJ : Metal Powder Industries Federation, 2007. p. 8.29-8.43. , May-2007

Articles in Peer-reviewed Journals	Park SJ, German RM. "Master curves based on time integration of thermal work in particulate materials." International Journal of Materials and Structural Integrity 2007 May 31;1(1-2):128-47. http://dx.doi.org/10.1504/IJMSI.2007.013868 , May-2007
Articles in Peer-reviewed Journals	Park SJ, Martin JM, Guo JF, Johnson JL, German RM. "Grain Growth Behavior of Tungsten Heavy Alloys Based on the Master Sintering Curve Concept." Metallurgical and Materials Transactions A, 2006 Nov; 37A(11): 3337-46. , Nov-2006
Articles in Peer-reviewed Journals	Park SJ, Martin JM, Guo JF, Johnson JL, German RM. "Densification Behavior of Tungsten Heavy Alloy Based on Master Sintering Curve Concept." Metallurgical and Materials Transactions A. 2006 Sep;37(9):2837-48. , Sep-2006
Dissertations and Theses	Campbell LJ. "Gravitational Effects on the Mechanical and Microstructural Properties of Tungsten Heavy Alloys." MS Thesis, Interdisciplinary Materials Program, Pennsylvania State University, State College, PA, December 2006. , Dec-2006
Papers from Meeting Proceedings	Martin JM, Johnson JL, German RM, Castro F. "Microstructural Evolution of Tungsten Heavy Alloys During Heating to the Sintering Temperature." Proceedings of the 2006 International Conference On Powder Metallurgy & Particulate Materials, San Diego, California, June 18–21, 2006. In: Advances in Powder Metallurgy and Particulate Materials - 2006. Princeton, NJ : Metal Powder Industries Federation, 2006, Part 5, pp. 43-58. , Jun-2006
Papers from Meeting Proceedings	Campbell LG, German RM. "Gravitational Effects on Mechanical and Microstructural Properties of Tungsten Heavy Alloys." 2006 International Conference on Tungsten, Refractory and Hardmetals IV, February 2006. In: Proceedings of the 2006 International Conference on Tungsten, Refractory and Hardmetals IV. A. Bose and R. Dowding (eds.), Princeton, NJ : Metal Powder Industries Federation, 2006, pp. 149-160. , Feb-2006
Papers from Meeting Proceedings	German RM, Park SJ, Chung SH, Johnson JL. "Prediction of Tungsten Heavy Alloy Density, Size, Shape, and Microstructure in Different Gravitational Environments." 2006 International Conference on Tungsten, Refractory and Hardmetals IV, February 2006. In: A. Bose and R. Dowding (eds.), Proceedings of the 2006 International Conference on Tungsten, Refractory and Hardmetals IV. Princeton, NJ : Metal Powder Industries Federation, 2006, pp. 71-82. , Feb-2006
Papers from Meeting Proceedings	Campbell LG, German RM. "Gravitational Effects on Microstructures in Liquid Phase Sintering." 2007 International Conference On Powder Metallurgy & Particulate Materials, Denver, Colorado, May 13-16, 2007. In: Advances in Powder Metallurgy & Particulate Materials—2007 Proceedings of the 2007 International Conference On Powder Metallurgy & Particulate Materials, May 13–16, 2007, Denver, Colorado. Princeton, NJ : Metal Powder Industries Federation, 2007. p. 8.10-8.28. , May-2007
Papers from Meeting Proceedings	Sethi G, Park SJ, Johnson JL, German RM. "Linking Homogenization and Densification in W-Ni-Cu Alloys Through Master Sintering Curve (MSC) Concepts. " International Conference On Powder Metallurgy & Particulate Materials, Denver, Colorado, May 13–16, 2007. In: Advances in Powder Metallurgy & Particulate Materials—2007 Proceedings of the 2007 International Conference On Powder Metallurgy & Particulate Materials, May 13–16, 2007, Denver, Colorado. Princeton, NJ : Metal Powder Industries Federation, 2007. p. 8.44-8.55. , May-2007